

CHORNOMORS'KE, UkraineWMO: **339240**Lat: **45.5217N**Lon: **32.7161E**Elev: **33**StdP: **14.68**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.8	17.6	7.1	7.9	14.8	12.0	10.2	19.4	36.1	30.5	31.6	28.4	12.4	70	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	86.3	72.3	83.7	71.8	81.7	71.2	75.6	82.3	74.1	81.0	72.7	79.7	11.6	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.3	124.0	80.7	71.7	117.2	79.4	70.2	111.1	77.9	39.1	82.2	37.7	80.9	36.4	80.0	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.6	24.1	22.1	7.7	93.3	7.1	4.3	2.6	96.3	-1.6	98.9	-5.6	101.3	-10.7	104.4		
(5)			7.1	77.8	6.9	2.2	2.1	79.4	-1.9	80.6	-5.8	81.8	-10.8	83.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	53.4	34.1	35.1	41.0	49.9	60.0	68.9	73.8	73.3	64.7	54.2	45.9	38.6	(6)
(7)		DBStd	15.55	8.84	8.89	6.37	5.76	5.62	4.50	3.91	4.43	5.32	7.18	7.80	8.85	(7)
(8)		HDD50	1836	492	417	284	70	2	0	0	0	0	41	168	362	(8)
(9)		HDD65	4975	956	836	744	452	173	14	0	2	68	339	572	819	(9)
(10)		CDD50	3083	1	1	6	68	313	567	737	722	441	172	46	9	(10)
(11)		CDD65	745	0	0	0	0	18	132	273	258	59	5	0	0	(11)
(12)		CDH74	4611	0	0	0	2	63	577	1846	1889	225	9	0	0	(12)
(13)		CDH80	883	0	0	0	0	6	75	357	422	21	2	0	0	(13)
(14)	Wind	WSAvg	9.9	11.4	11.7	11.5	9.7	8.1	8.3	8.5	9.2	10.0	10.9	11.5	(14)	
(15)	Precipitation	PrecAvg	15.7	1.3	1.3	1.0	1.1	1.0	1.4	1.3	1.5	1.7	1.2	1.4	1.8	(15)
(16)		PrecMax	22.9	3.3	3.0	2.1	2.8	2.7	5.4	4.7	6.7	6.8	4.2	4.7	4.1	(16)
(17)		PrecMin	7.6	0.1	0.3	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.2	(17)
(18)		PrecStd	3.9	0.9	0.6	0.6	0.7	0.8	1.1	1.4	1.6	1.7	1.0	0.9	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.5	55.9	60.6	70.6	79.4	86.1	90.7	91.5	82.7	74.2	66.3	58.9	(19)
(20)			MCWB	49.1	49.2	50.7	56.8	64.0	70.4	70.9	70.9	66.1	64.8	59.2	53.2	(20)
(21)		2%	DB	50.3	51.9	55.5	65.2	75.2	82.0	86.0	86.8	79.1	70.3	62.7	55.6	(21)
(22)			MCWB	47.1	47.4	48.3	54.4	64.7	70.2	72.0	73.2	67.0	63.1	57.4	52.1	(22)
(23)		5%	DB	48.2	49.1	52.4	61.7	72.1	79.2	83.5	84.3	76.5	67.3	59.7	52.9	(23)
(24)			MCWB	45.7	45.8	46.8	53.2	63.3	69.4	72.4	72.2	66.8	61.4	55.7	49.9	(24)
(25)		10%	DB	46.1	46.5	49.7	59.0	69.6	77.3	81.6	82.0	74.2	64.8	57.1	50.7	(25)
(26)			MCWB	43.8	44.0	45.4	51.8	62.1	68.5	71.8	71.4	65.3	59.8	53.8	47.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.6	50.2	51.5	58.9	68.6	74.5	77.0	79.2	72.5	66.8	60.4	55.1	(27)
(28)			MCDWB	52.6	54.0	58.6	68.5	74.9	81.1	83.2	85.4	78.0	71.8	64.7	57.8	(28)
(29)		2%	WB	47.4	48.1	49.2	56.1	66.5	72.5	75.5	76.1	70.1	64.5	57.9	52.1	(29)
(30)			MCDWB	49.9	51.3	54.1	62.9	72.4	78.7	81.9	83.2	75.9	68.9	61.6	54.9	(30)
(31)		5%	WB	45.9	45.9	47.5	54.1	64.7	71.0	74.3	74.4	68.4	62.5	56.0	50.2	(31)
(32)			MCDWB	48.1	48.5	51.6	59.8	70.4	77.2	81.0	81.7	74.0	66.7	59.2	52.8	(32)
(33)		10%	WB	44.0	44.1	45.9	52.7	63.0	69.7	73.0	72.8	66.6	60.3	54.2	48.2	(33)
(34)			MCDWB	46.0	46.5	49.3	57.9	68.3	75.9	80.0	80.1	72.3	64.1	57.1	50.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.8	9.5	9.9	12.6	13.0	12.1	12.5	13.5	13.3	12.2	10.3	8.6	(35)
(36)			MCDBR	10.6	12.2	14.4	17.3	17.7	15.8	16.4	17.9	17.3	14.6	12.7	9.8	(36)
(37)			MCWBR	8.5	8.7	8.7	9.4	9.1	8.3	8.7	9.5	9.6	9.2	8.6	7.9	(37)
(38)		5% WB	MCDBR	10.4	11.2	13.0	15.0	14.9	13.5	13.8	14.5	14.5	12.8	11.7	9.3	(38)
(39)			MCWBR	8.8	8.5	8.6	9.2	8.8	8.5	8.5	9.4	9.6	9.0	8.6	8.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.350	0.376	0.401	0.439	0.417	0.418	0.438	0.442	0.413	0.381	0.362	0.348	(40)	
(41)		taud	2.327	2.257	2.219	2.139	2.300	2.342	2.292	2.263	2.312	2.372	2.343	2.328	(41)	
(42)		Ebn at Noon	233	249	260	260	271	270	263	257	254	246	226	219	(42)	
(43)		Edh at Noon	25	33	39	46	40	39	40	40	35	29	25	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	350	634	1056	1610	2031	2146	2155	1881	1370	847	441	289	(44)	
(45)		RadStd	49	77	101	120	127	122	92	103	126	81	51	40	(45)	